

```

#include <stdio.h>

#define SIZE 100
#define INVALID 0
#define DOWN 2
#define RIGHT 6
#define LEFT 4
#define UP 8

int isSafe(int (*arr)[SIZE], int n, int m, int col, int row)
{
    return (col >= 0 && col < m && row >= 0 && row < n && arr[row][col]);
}

int getDirection(int (*arr)[SIZE], int n, int m, int d, int row, int col)
{
    if ((d == RIGHT || d == UP && !isSafe(arr, n, m, col, row - 1))
        && isSafe(arr, n, m, col + 1, row))
        return RIGHT;
    else if ((d == DOWN || d == RIGHT && (!isSafe(arr, n, m, col + 1, row)))
        && isSafe(arr, n, m, col, row + 1))
        return DOWN;
    else if ((d == UP || d == LEFT && !isSafe(arr, n, m, col - 1, row))
        && isSafe(arr, n, m, col, row - 1))
        return UP;
    else if ((d == LEFT || d == DOWN && !isSafe(arr, n, m, col, row + 1))
        && isSafe(arr, n, m, col - 1, row))
        return LEFT;
    else
        return INVALID;
}

int main()
{
    int arr[SIZE][SIZE] = {0}, i, j, row, col, direction = RIGHT;

    scanf("%d %d", &row, &col);

    for (i = 0; i < row; i++)
        for (j = 0; j < col; j++)
            scanf("%d", &arr[i][j]);

    i = 0, j = 0;
    while(1)
    {
        direction = getDirection(arr, row, col, direction, i, j);
        printf("%d ", arr[i][j]);
        arr[i][j] = 0;
        if (direction == RIGHT)
            j++;
        else if (direction == DOWN)
            i++;
        else if (direction == LEFT)
            j--;
        else if (direction == UP)
            i--;
        else
            break;
    }
}

```

}

}